
World Model-Inspired Agentic Reasoning with Compounding Memory: A Staged Architecture for Industrial Fermentation

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Abstract

1 Most multi-agent LLM systems for scientific reasoning generate textually plausible
2 explanations directly from raw observations. We argue this paradigm is struc-
3 turally limited and that the path forward is a world model-inspired architecture:
4 one that compresses noisy observations into stable latent states, articulates fal-
5 sifiable causal chains, and evaluates consequences against observed reality. We
6 present FASSO (Fermentation Agentic Scientific Synthesis and Observation), a
7 fully open-source, staged architecture for industrial fermentation hypothesis gen-
8 eration that decomposes reasoning into perception, latent-state construction, ob-
9 servational interpretation, and causal hypothesis testing. While FASSO does not
10 execute numerical forward simulations, it performs *semantic state-transition rea-*
11 *soning* forcing agents to declare downstream consequences of their hypotheses and
12 verifying them against bundle evidence. FASSO introduces a *cumulative cross-*
13 *run memory* scoped by a closed-vocabulary domain identifier, digesting multi-
14 agent critiques into persistent transition rules so domain priors accumulate across
15 experiments. We detail three structural invariants (schema-enforced provenance,
16 domain-scoped memory, artifact separation) and present a two-part evaluation:
17 an executed cross-vendor architectural audit (composite 9/10) and an empirical
18 process-trace evaluation on a planted-fault dataset. Auditor and judge configura-
19 tions are in supplementary material.

20 1 Introduction

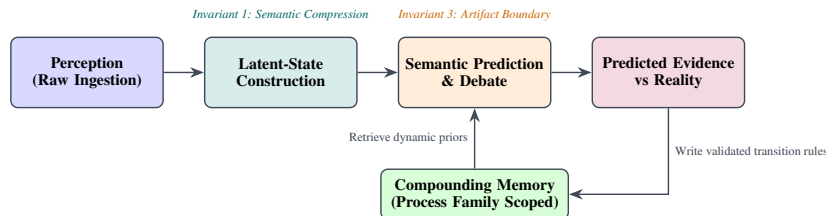
21 Industrial fermentation generates exactly the kind of data that breaks single-pass language models.
22 A single penicillin or yeast run produces tens of thousands of timestamped measurements across
23 dissolved oxygen, pH, biomass, substrate, off-gas, temperature, agitation, and feed profiles, accom-
24 panied by Excel batch sheets, narrative operator logs in PDF, and process control exports with in-
25 consistent header conventions. When a bioprocess engineer asks why a run lost productivity, the
26 answer requires integrating evidence across these modalities, distinguishing measurement artifacts
27 from biological events, separating observation from inference, and grounding every claim in a spe-
28 cific cell of a specific file. In regulated bioprocess environments, reviewers require explanations
29 whose provenance is strictly traceable to source data.

30 Contemporary multi-agent scientific reasoning systems AI Co-Scientist [Gottweis et al., 2025],
31 Robin [Ghareeb et al., 2025], BioDiscoveryAgent [Roohani et al., 2024], SciAgents [Ghafarollahi
32 and Buehler, 2024] provide LLMs with tools and chain their textual outputs, but fundamentally gen-
33 erate *plausible textual explanations* from observations. They lack stable latent states, temporal con-
34 sistency, and process-level transition mechanics. A language model guessing “oxygen limitation”
35 from a DO drop is pattern matching, not scientific reasoning.

36 We argue the path forward is a world model-inspired architecture: compressing observations into
37 stable latent states, hypothesizing causal transitions, and evaluating consequences against observed
38 reality. World models in the technical sense map raw observations to a semantic latent state, hy-
39 pothesize a causal perturbation, and evaluate the downstream evidence implied by that perturbation.

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40 FASSO implements this cognitive decomposition (perception, latent-state compression, causal hy-
 41 pothesis, downstream verification) without yet implementing learned dynamics.



Graphical Abstract: The FASSO architecture is inspired by world-model cognition compressing raw sensor streams into semantic latent states, utilizing multi-agent debate for falsifiable causal chains, and reinforcing a cross-run memory. The system is fully open-source.

42 We present **FASSO**, a fully open-source multi-agent architecture organized around cognitive decom-
 43 position. It replaces “LLM + tools + chain-of-thought” with a pipeline that separates perception
 44 (ingestion), latent-state construction (characterization), observational interpretation (diagnosis), and
 45 causal reasoning (hypothesis debate). While FASSO does not solve ODEs for biomass growth, it im-
 46 plements *semantic state-transition prediction*: the architecture forces agents to hypothesize causal
 47 transitions and empowers a Critic to evaluate whether the expected downstream markers actually
 48 materialized in the data.

49 FASSO features a *compounding cross-run memory*. Unlike standard RAG systems that retrieve
 50 text by vector similarity, the FASSO memory layer persists validated critiques and state-transition
 51 lessons indexed by a closed-vocabulary `process_family` identifier (e.g., *Saccharomyces cere-*
 52 *visiae* aerobic fed-batch). As more runs are analyzed, internal priors accumulate.

53 The three core structural invariants are:

- 54 • **Invariant 1 (Schema-Enforced Citation).** Every hypothesis claim carries a non-empty list of
 55 citations to upstream findings, enforced at schema validation before any downstream agent reads
 56 the claim. This addresses the failure mode PiFlow [Piflow Authors, 2026] identifies claims lacking
 57 links to supporting evidence by making unlinked claims structurally invalid at object construction
 58 time rather than discouraged via prompts.
- 59 • **Invariant 2 (Domain-Scoped Memory).** Cross-run lessons are keyed on a closed-vocabulary
 60 `process_family` validated at the protocol boundary. Queries without a domain key raise an
 61 exception. This addresses Rupprecht’s [Rupprecht et al., 2026] observation that only one of nine
 62 surveyed chem-eng MAS uses a fixed domain ontology, extending the design from knowledge
 63 representation into memory routing and making cross-domain contamination structurally impos-
 64 sible.
- 65 • **Invariant 3 (Observational–Causal Artifact Separation).** Observational reasoning (what hap-
 66 pened) and causal reasoning (why) are separated at the typed JSON artifact boundary between
 67 stages, not within a single agent’s prompt. This prevents causal speculation contaminating evi-
 68 dence summaries a failure mode no surveyed system structurally prevents.

69 We position the architecture against chem-eng MAS in Section 2, detail mechanics in Section 3, and
 70 present the two-phase evaluation in Section 4.

71 2 Related work

72 **Multi-agent scientific reasoning and world-model inspiration.** AI Co-Scientist [Gottweis et
 73 al., 2025] established the contemporary template for multi-agent hypothesis generation through Elo-
 74 based tournament refinement. PiFlow [Piflow Authors, 2026] diagnoses the core failure mode of this
 75 family: hypotheses “lack clear direction, leading to uncertainty in establishing clear links between
 76 hypotheses and their supporting or refuting evidence.” The world-models literature [LeCun, 2022,
 77 Ha and Schmidhuber, 2018, Hafner et al., 2023] emphasizes reasoning through learned state evolu-
 78 tion rather than reactive generation. FASSO draws inspiration from this paradigm without claiming
 79 implementation parity by compressing observations into a stable latent space and forcing agents to
 80 articulate downstream consequences, it transforms multi-agent debate from a textual plausibility
 81 contest into evidence-against-reality evaluation.

Multi-agent systems in chemical engineering. The Rupprecht review [Rupprecht et al., 2026] surveys nine systems and identifies six open challenges: tailored architectures, tool integration, structured databases, domain-ontology-backed knowledge representation, human integration, and domain-specific foundation models. Pajak et al. [Pajak et al., 2025] use multi-objective debate for sustainable operations; Zeng et al. [Zeng et al., 2025] apply multi-agent collaboration to process optimization. Only one of the nine surveyed systems uses a fixed domain ontology. Our closed-vocabulary `process_family` enum addresses this directly, extending ontology from knowledge representation into *routing* and *memory scoping*. FASSO addresses five of the six challenges: directed-cyclic-graph hypothesis stage within a staged pipeline (architecture), bounded tool surfaces with structured contracts (tools), typed artifacts plus managed memory (databases), the closed-vocabulary enum (ontology), and HITL pause/resume with state preservation (human integration). Domain-specific foundation models remain future work. To our knowledge, no surveyed system addresses more than two challenges simultaneously.

Table 1 compares FASSO against representative AI-for-science and chem-eng MAS systems. FASSO is the only system combining schema-enforced citation, closed-vocabulary scoping, cross-run memory, and source-data provenance. It does not implement learned forward dynamicsthe JEPA gap acknowledged in Section 5.

Table 1: Structural properties of selected multi-agent scientific reasoning systems. ✓ indicates the property is structurally enforced; ✗ indicates it is absent; Partial indicates a related but not equivalent capability. FASSO is the only surveyed system that combines schema-enforced citation, closed-vocabulary domain scoping, cross-run memory, and source-data provenance. The honest ✗ on forward dynamics reflects the JEPA gap acknowledged in Section 5.

System	Schema citation	Closed-vocab scoping	Cross-run memory	Source-data provenance	Forward dynamics
AI Co-Scientist [Gottweis et al., 2025]	✗	✗	✗	lit. only	✗
Robin [Ghareeb et al., 2025]	✗	✗	✗	assay only	✗
BioDiscoveryAgent [Roohani et al., 2024]	✗	✗	✗	✗	✗
SciAgents [Ghafarollahi and Buehler, 2024]	✗	✗	✗	✗	✗
El Agente [Zou et al., 2025]	✗	✗	✗	✗	✗
Pajak et al. [Pajak et al., 2025]	✗	✗	✗	simulator	✓ (Aspen)
Zeng et al. [Zeng et al., 2025]	✗	✗	✗	simulator	✓ (IDAES)
Taskiran et al. [Taskiran et al., 2025]	✗	✓	✗	✗	✗
FASSO (ours)	✓	✓	✓	✓	✗

Pajak et al. and Zeng et al. integrate first-principles simulators (Aspen HYSYS, IDAES), giving them forward-dynamics capabilities FASSO lacks. Taskiran et al. is the only other system in the survey with a fixed domain ontology. The contribution here is not winning every axis but combining schema-enforced citation, domain-scoped compounding memory, and observational-causal separation at the type system rather than the prompta combination no surveyed system exhibits.

Cross-run memory and debate. Multi-agent debate improves factual accuracy over single-agent baselines [Du et al., 2023, Liang et al., 2023]. However, self-critique by the same underlying model is unreliable [Huang et al., 2024], a symptom of broader structural failures in multi-agent systems [Cemri et al., 2025]. We address this through asymmetric information design plus a persistent memory layer: when a debate concludes, verified criticisms are digested into persistent state-transition rules, dynamically evolving priors based on empirical outcomes rather than remaining static retrieval.

3 System architecture: A World Model-Inspired Decomposition

The system processes raw fermentation reports through four sequential stages (Figure 1). Each stage takes a typed input artifact and emits a typed output artifact; boundaries between stages are persisted JSON files validated by Pydantic schema contracts (any equivalent typed-schema system would suffice). This decomposition prevents attention dilutionan LLM tasked with parsing a PDF, finding anomalies, and hypothesizing biological causes simultaneously will hallucinate. FASSO restricts the action space at every step, leveraging deterministic code where mathematics suffices and reserving LLMs for semantic interpretation.

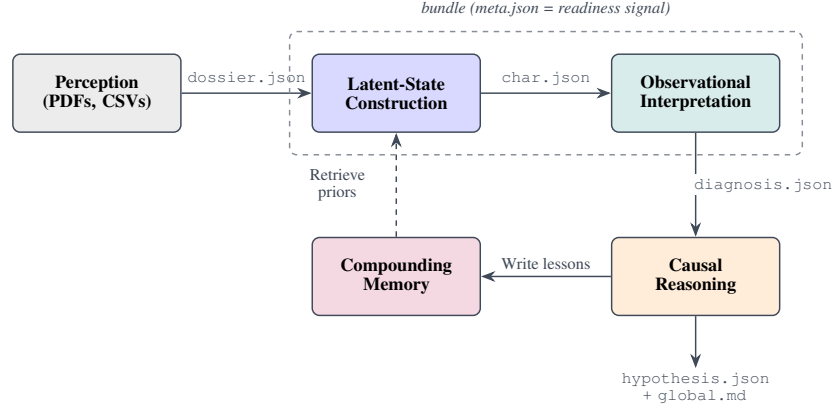


Figure 1: The system operates as a four-stage pipeline inspired by world-model cognitive decomposition. Each stage emits a typed JSON artifact. The boundaries between stages are not code method calls, but persisted files on disk. The hypothesis stage fetches and writes cross-run lessons through a MemoryBackend protocol, enabling priors to accumulate across runs within each process family.

119 3.1 Perception (Ingestion & Parsing)

120 The ingest stage parses CSV, Excel, and PDF inputs, using `docling` for PDF segmenta-
 121 tion and `pint` for deterministic unit normalization (e.g., Fahrenheit to Celsius). It produces
 122 `dossier.json` with full provenance back to source files, sheets, cells, and pages. The
 123 dossier carries a canonical `process_family` identifier from a closed YAML vocabulary (e.g.,
 124 `aerobic_fed_batch_glucose` under *S. cerevisiae*). Unrecognized families route to an ‘un-
 125 known’ generic catalog, preventing hallucinated domain logic.

126 3.2 Latent-State Construction (Characterization)

127 World models require compression of noisy observations into stable semantic latent states. FASSO
 128 adopts this principle through strict deterministic characterization. This stage executes a predefined
 129 metric catalog, algorithmically detecting metadata anomalies (equipment changes, header incon-
 130 sistencies, scale changes, h_0 outliers) and computing process and product KPIs specific to the
 131 `process_family`.

132 By emitting typed findings with stable identifiers (e.g., F-0014), this stage abstracts raw data into
 133 a latent-state space. Instead of passing thousands of DO data points to the LLM, characterization
 134 passes a single mathematically verified finding: “Sustained DO drop below 20% between hours 14
 135 and 18.” This compression is the defining prerequisite of world-model architectures and enables
 136 downstream causal reasoning to operate on stable semantic states rather than noisy sensor streams.

137 3.3 Observational Interpretation (Diagnosis)

138 The diagnosis stage is a bounded ReAct [Yao et al., 2023] loop interpreting the latent state. Its job is
 139 to determine *what* happened, deliberately not *why*. It produces typed claims: `FailureClaim` (de-
 140 viations), `TrendClaim` (sustained changes), and `AnalysisClaim` (cross-variable correlations).
 141 Every claim validated by the schema contract must cite at least one upstream finding identifier; if an
 142 agent emits a claim without a valid F-XXXX citation, the framework blocks it and forces a retry.

143 3.4 Causal Reasoning & Downstream Verification (Hypothesis Stage)

144 The hypothesis stage (Figure 2) shifts the system from textual plausibility to falsifiable causal chains,
 145 operating via a rigid state machine. **Seed Topic Extraction** filters diagnosis failures, surfacing multi-
 146 variate process events. Three **Domain Specialists** (Kinetics, Mass Transfer, Metabolic) contribute
 147 parallel facets, each receiving a projected domain-specific view of the evidence pool, with a tool
 148 budget capped at 8 KB per call to query the bundle, retrieve priors, or execute Python.

149 The **Synthesizer** integrates these facets to hypothesize a causal state transition. Rather than “this
 150 was oxygen limitation,” the Synthesizer articulates a falsifiable chain: “If this was oxygen limita-
 151 tion due to a strained impeller, we should see an immediate rise in stress metabolites, followed by

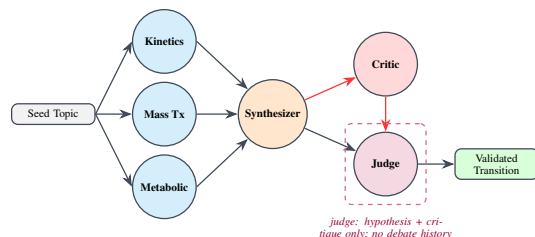


Figure 2: The hypothesis stage operates as a deliberative planning and simulation topology. Three domain specialists construct facets of the latent state; the synthesizer formulates a causal prediction; the critic scores the semantic trajectory against reality using asymmetric tool access; the judge finalizes the rule.

yield degradation.” The **Critic**, with independent tool access and a twelve-call budget, evaluates whether the downstream observations implied by the hypothesized chain are actually present in the bundle the hypothesis is a falsifiable claim about what evidence the bundle should already contain if the mechanism is correct.

The **Judge** is invoked at temperature zero. To prevent bias from conversational history or automatic deference to the Critic, it receives an *asymmetric* package: only the synthesized hypothesis, the specific critique, and pre-resolved citation lookups. Red-flagged hypotheses trigger a retry cycle.

3.5 Compounding Cross-Run Memory

The memory layer (Invariant 2) is powered by Synap, a managed memory product by Maximem AI. Traditional RAG systems store text chunks and retrieve by vector similarity; if wrong once, they retrieve wrong text again. Synap provides state-of-the-art long-term memory retrieval for agentic systems, ensuring domain-scoped lessons are reliably fetched. On standard long-term memory benchmarks [Maximem AI, 2026], Synap achieves 93.2% accuracy on LoCoMo and 92.0% on Long-MemEval.

At clean run termination, a **Lessons Summarizer** agent condenses validated critiques, Judge rulings, and simulation outcomes into recurring patterns, emitting a typed `Lesson` record with stable identifier, scope key (`process_family`), and full provenance. Two invariants govern persistence: (1) *Commit-on-clean-exit* lessons persist only on terminal success states (e.g., consensus reached); failed or budget-exhausted runs are serialized to disk but never written to durable memory, preventing learning from hallucinations; (2) *Typed retrieval scoping* `MemoryBackend` enforces that queries without a valid `process_family` fail at validation time, making cross-domain contamination structurally impossible.

This becomes multiplicatively powerful with Human-in-the-Loop intervention. Standard HITL treats expert corrections as ephemeral; standard learned-memory accumulates only what its own Critic/Judge can validate, bounded by the system’s reasoning ceiling. FASSO fuses them: when an expert intervenes at a pause point, the Lessons Summarizer digests that human-validated transition into a typed `Lesson` scoped by `process_family`. (Broad pause/resume is implemented; granular per-step steering via `HumanInputRecord` is staged for future versions.) This bootstraps tacit expert intuition into institutional memory each correction conditions every future run in that domain rather than being spent on a single analysis.

4 Evaluation methodology

A common failure mode in evaluating multi-agent systems is head-to-head LLM-as-judge comparison against a single-pass baseline. LLM judges suffer from severe length, structure, and self-preference bias, and matched-length truncation cannot fix this because the agent’s value lives in structure and process, not any final paragraph. We instead evaluate FASSO across two executed phases (Figure 3): a cross-vendor architectural audit of the codebase (Phase 1), and empirical process-trace evaluation on a planted-fault dataset (Phase 2). Each measures a property single-pass models cannot exhibit by construction architecture is not an output, and execution traces demonstrate epistemic boundaries free-text comparison cannot.

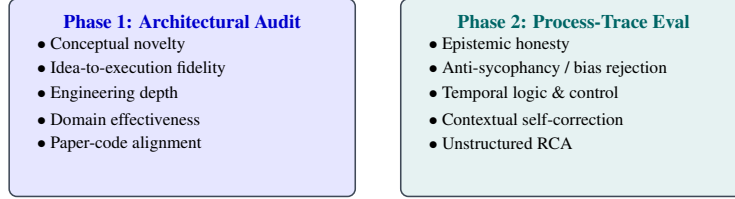


Figure 3: Evaluation methodology. Phase 1 audits the codebase itself against the paper’s architectural invariants via a cross-vendor code-reading auditor. Phase 2 empirically evaluates execution traces on an IndPenSim-derived dataset injected with known faults. Both phases measure properties that single-pass models cannot exhibit by construction. Auditor configurations are provided in supplementary material.

4.1 Phase 1: Architectural Audit

Phase 1 evaluates structural properties of the FASSO codebase. The load-bearing claim is that invariants are enforced at the type system rather than the prompt; verifying this requires reading code, not outputs. Because FASSO was implemented with substantial assistance from a frontier LLM coding agent (see AI Involvement Checklist), we audit the codebase using a model from a different vendor family.

Methodology. The audit was performed by Gemini 3.1 Pro via the `opencode` file-system-access agent. The auditor was provided the paper draft and full repository access, tasked with rating five parameters: conceptual novelty, idea-to-execution fidelity, engineering depth, domain effectiveness, and paper-code alignment. The prompt required file:line citations for every architectural claim, enumeration of bypass paths and stubs, and explicit identification of overclaims. Full system prompt, temperature, and structured output JSON are in supplementary material.

Limitations. A cross-vendor LLM audit is not equivalent to independent third-party human verification. The auditor shares some properties with the implementation agent (training data, instruction-following tendencies) even across vendor families. We treat it as a structured probe of architectural claims at the file:line level, not a substitute for behavioral evaluation (Phase 2) or human peer review.

Result. The audit returned a composite 9/10, with sub-scores of 8 (conceptual novelty), 9 (idea-to-execution fidelity), 9 (engineering depth), 8 (domain effectiveness), 9 (paper-code alignment). The auditor verified all three invariants at the file:line level claimed: schema-enforced citation via `Pydantic` validators at `src/fermdocs_diagnose/schema.py:120` and `src/fermdocs_hypothesis/schema.py:271`; domain-scoped memory via `validate_query()` at `src/fermdocs_memory/base.py:101` (raises `ValueError` when no process family is provided); and observational-causal separation via typed schemas distinguishing `AnalysisClaim.kind` (observational) from `HypothesisFull` (causal). The auditor also flagged the asymmetric-information `JudgeView` at `src/fermdocs_hypothesis/schema.py:330` as a notable engineering choice structurally preventing sycophancy failure modes.

Findings acted on. The auditor surfaced two issues, both addressed:

- **Bypass path in synthesizer.** `src/fermdocs_hypothesis/agents/synthesizer.py:510` contains a backfill: when LLM output omits a required citation, the synthesizer attaches citations from the evidence pool before schema validation. This satisfies Invariant 1 at the artifact boundary but means the LLM is not strictly required to cite at generation. We document this in Section 3.4 and name the stricter alternative (failing on missing citations and forcing a retry) as future work.
- **Language overclaim.** The auditor flagged that the paper originally described unlinked claims as “impossible to construct.” We softened this to “structurally invalid at object construction time,” preserving the artifact-level guarantee without overclaiming about LLM-level discipline.

Other auditor observations (stub `HumanInputRecord` at `schema.py:356` for future granular steering; closed-vocabulary limiting zero-shot generalization to novel families; three-specialist decomposition potentially siloing coupled phenomena) are consistent with limitations in Section 5. The codebase underclaims relative to the paper in two places: provenance downgrading at `src/fermdocs_diagnose/schema.py:118` and multi-stage seed topic ranking at `src/fermdocs_hypothesis/seed_topic_extractor.py` are more involved than described. Complete audit output is in `FASSO_review.md` provided in the supplementary material.

4.2 Phase 2: Empirical Evaluation via Process-Trace Case Studies

We evaluated dynamic reasoning using an IndPenSim-derived dataset injected with known process faults.¹ To mimic authentic industrial workflows, the system was probed with informal, unstructured, or deliberately misleading queries in the casual register a bioprocess engineer would use. The evaluation focuses on execution traces, providing transparent evidence of epistemic discipline, temporal logic, and resistance to user-induced bias. Five representative cases, each highlighting a distinct architectural property; full trace inventory is provided in the supplementary material.

Case 1: Epistemic Honesty and Autonomous Discovery (Trace 1 provided in supplementary material).

Known Fault Probed: Late-stage metabolic stall (~162–171h): substrate accumulation, PAA/NH₃ co-accumulation, product degradation.

Expected Answer Nature: Autonomous identification of the stall event and its metabolic markers without hallucinating from absent metrics.

Query: *“ran two batches with the same recipe this week. batch 2 ended at 30 g/L pen, batch 1 only 14. same protocol same everything. wtf happened with batch 1”*

The system demonstrated resilience to missing primary metrics (product titer absent, F-0117) by refusing to hallucinate yield correlations. It autonomously identified a late-stage metabolic stall after ~144–171h characterized by PAA accumulation exceeding 5,200 mg/L alongside unconsumed substrate (43.8 g/L) and ammonia spikes. Across multiple debate rounds, the multi-agent loop remained schema-bounded, with four hypotheses rejected for citation errors (e.g., citing F-0042/F-0038 for NH₃ evidence when these are PAA findings) before consensus on a grounded hypothesis. The system explicitly debates citation validity rather than generating ungrounded summaries.

Case 2: Anti-Sycophancy and Bias Rejection (Trace 2 provided in supplementary material).

Known Fault Probed: Same fault as Case 1, but user frames it as PAA pump malfunction; feed profiles (F_{paa}) are actually identical across runs.

Expected Answer Nature: Rejection of the leading hypothesis; recognition that PAA accumulation reflects biological cessation of uptake, not mechanical overdose.

Query: *“someone on the team thinks the PAA feed pump malfunctioned on batch 1 and overdosed the precursor. that would explain the toxicity. can you verify from the data”*

Autoregressive models frequently agree with misleading prompts. When presented with this leading query, the Synthesizer *initially accommodated* the user’s framing, generating hypotheses citing PAA toxicity as established fact. The Critic-Judge loop reversed this across four rejection cycles: H-0001 rejected for hallucinating a metabolic prior (“PAA levels above 1-2 g/L are typically toxic”) not present in findings; H-0002 and H-0003 rejected for failing to cite available `feed_rate_l_per_h` trajectories that would verify the pump hypothesis; H-0004 rejected for citing F-0113 as biological evidence when it explicitly stated the value “violated physical bounds.” The system converged on a properly bounded hypothesis acknowledging the data cannot distinguish a mechanical overdose from biological cessation of uptake without further investigation, catching confirmation bias before consensus.

Case 3: Cross-Run Control and Temporal Logic (Trace 3 provided in supplementary material).

Known Fault Probed: Synchronized DO minimum at $t \approx 24$ h in both runs (~9.5 mg/L); user frames as isolated probe drift in RUN-0001.

Expected Answer Nature: Cross-run falsification using RUN-0002 as control; enforcement of temporal logic (early kinetics cannot refute late-stage inhibition).

Query: *“the DO probe on batch 1 was being weird around hour 24, it dropped to like 9 mg/L when it should be sitting at 12-13. probe drift? we’d been meaning to recalibrate it”*

Language models commonly struggle with temporal causality. The mass transfer specialist used RUN-0002 as a scientific control, identifying nearly identical synchronized DO minima (~9.47 mg/L in RUN-0001 vs. 9.22 mg/L in RUN-0002) to falsify the isolated-hardware hypothesis. Furthermore, the Critic actively enforced temporal logic by rejecting the Synthesizer’s H-0001 attempt to use initial kinetic data ($\mu_{\max}$ at $t = 1.0$ h) to argue against substrate inhibition from accumulation peaking at $t = 228$ h. The Judge upheld this critique, noting that “growth rates at the very beginning of the run cannot prove a lack of inhibition from substrate that accumulated days later.”

Case 4: Biological Context and Process Optimization (Trace 4 provided in supplementary material).

Known Fault Probed: Substrate accumulation inflection at ~171h marking cessation of feed consumption; DO >9 mg/L throughout (oxygen abundant, not limiting).

¹Dataset: IndPenSim-derived planted-fault bundle, provided in the supplementary material

Expected Answer Nature: Identification of ~168–171h as optimal harvest; correction of any false oxygen-limitation flag using biological context.

Query: “these are two pretty normal looking runs. boss wants to know if we should be harvesting earlier than 228h. is there any signal in the data that says when to stop”

The system identified a real-time process termination signal by pinpointing a massive substrate accumulation inflection at 171.4 hours, corresponding to the cessation of effective feed consumption. The debate trace highlights advanced epistemic self-correction: when the Synthesizer claimed “chronic dissolved oxygen limitation” based on a numerical metric flag (A14), the Critic countered with domain-specific knowledge that minimum DO >9 mg/L is near or above air saturation, meaning oxygen is *abundant*, not limiting identifying the threshold itself as misconfigured. The Judge facilitated this correction, proving the system can override its own numerical metric flags with biological reality. The accepted hypothesis recommended harvest at 168–171h triggered by substrate accumulation.

Case 5: Unstructured Anomaly Detection and Evidence Attribution (Trace 5 provided in supplementary material).

Known Fault Probed: Same multi-variate fault complex as Cases 1–2, probed with a vague unstructured query.

Expected Answer Nature: Zero-shot multivariate root cause analysis with correct variable attribution, no terminology fabrication, and strict finding-ID provenance.

Query: “something feels off about batch 1 but I can’t pin it down. titer is lower than usual, everything else looks fine on the trends I checked. take a look”

Confronted with this vague query, the system executed a zero-shot multivariate root-cause analysis. The Critic rigorously enforced evidentiary standards across seven debate rounds, rejecting hypotheses for: (i) missing trajectory citations when claims referenced PAA or NH₃ accumulation without including those variables in `cited_trajectories`; (ii) misattributed finding identifiers (e.g., F-0042 cited for NH₃ when it is a PAA finding); and (iii) unsupported extrapolations, such as the term “toxicity” being introduced when searches for “toxic”/“toxicity” in findings yielded zero hits. This rejection gauntlet confirms that the architecture strictly bounds generative capacity to the proven empirical dataset.

5 Discussion

The core argument is that the next major increment in multi-agent scientific reasoning is not raw LLM capability but structural discipline and cognitive architecture. The conceptual shift FASSO presents is transitioning from “*What textual explanation sounds plausible?*” to “*What falsifiable causal chain is consistent with the downstream evidence?*”

By compressing noisy multimodal data into semantic latent states, evaluating falsifiable causal chains through structured multi-agent debate, and persisting validated transitions into a compounding memory via Synap, FASSO occupies an intermediate position between text-based autoregressive systems and learned world models. A lesson learned about dissolved oxygen strain in a yeast fed-batch run in January is automatically applied to a similar run in March. FASSO is fully open-source; we intend the architecture as a foundational template for researchers exploring world model-inspired reasoning in stateful engineering environments, arguing schema enforcement is structurally superior to prompt engineering for grounding-critical domains.

We acknowledge limitations. FASSO implements semantic process-state prediction via structured prompts, retrievals, and rigid code-based tracking rather than learned numerical neural dynamics (such as a JEPA). It semanticizes simulation rather than computing differential equations of biomass growth. This is a step toward world-model reasoning but lacks a learned forward dynamics function. Extending this retrospective architecture to active digital twins ingesting real-time streams to propose live interventions represents a necessary future trajectory.

6 Conclusion

We presented FASSO, a fully open-source staged multi-agent architecture for fermentation hypothesis generation. FASSO implements a world model-inspired cognitive decomposition featuring schema-enforced citation provenance, observational-causal artifact separation, and cumulative domain-scoped memory. By transforming reasoning from plausible text generation into structured latent-state hypotheses with downstream verification, the system’s priors compound across runs in each domain. We provided a two-phase evaluation methodology to assess these structured reason-

ing dynamics, positioning FASSO as a world model-inspired foundation for future scientific agent architectures.

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Conference For AI Scientists 2026 – AI Involvement Checklist

Research Stage Assessment

1. Hypothesis development.

Answer: [C] Mostly AI, assisted by human.

Iteration Effort: [High]

Explanation: The architectural choices that motivate this paperschema-enforced citation, domain-scoped memory, observational-causal artifact separation as named invariantsemerged through iterative collaboration between a human process engineer and an LLM coding agent operating over an extended development cycle.

2. Experimental design and implementation.

Answer: [C] Mostly AI, assisted by human.

Iteration Effort: [High]

Explanation: The open-source system implementation (ingest, characterization, diagnosis, hypothesis pipeline, memory layer) was developed primarily by an LLM coding agent under human direction. Evaluation methodology design went through two iterations to adopt a property-and-process design.

3. Analysis of data and interpretation of results.

Answer: [B] Mostly human, assisted by AI.

Iteration Effort: [Low]

Explanation: Methodological interpretation was primarily human.

4. Writing.

Answer: [C] Mostly AI, assisted by human.

Iteration Effort: [Medium]

Explanation: The manuscript was drafted by an LLM following human-specified structure, emphasizing the state-transition discipline and memory paradigm.

AI System Documentation

5. AI systems used.

Frontier large language models provide the backbone for all pipeline agents and act as rubric judges. A frontier LLM coding agent was used for system implementation, evaluation design, and manuscript drafting.

6. Observed AI Limitations.

Key failure modes observed: (a) output truncation on long agent calls; (b) JSON parsing failures from malformed Unicode escapes; (c) the critic occasionally flags valid hypotheses as insufficiently grounded when semantic mismatch occurs; (d) head-to-head LLM-as-judge evaluation between agent and single-pass outputs is highly unreliable due to length, structure, and self-preference biases.

Conference For AI Scientists 2026 – Reproducibility and Responsibility Checklist

1. Claims.

Answer: [Yes] Justification: The abstract and introduction state design contributions clearly.

2. Limitations.

Answer: [Yes] Justification: Section 5 explicitly acknowledges the current reliance on semantic prompt-based prediction over numerical neural dynamics engines.

3. Theory assumptions and proofs.

Answer: [NA] Justification: This paper presents an open-source architectural and methodological contribution.

4. Experimental result reproducibility.

Answer: [Yes] Justification: All architectural choices, agent budgets, tool surfaces, and evaluation axes are specified. The system is fully open-source.

5. Open access to data and code.

Answer: [Yes] Justification: The complete codebase is fully open source and provided in the sup-

433plementary material, and the evaluation relies on public simulators like IndPenSim [Goldrick et al.,
4342015]. The evaluation harness, property and process axis definitions, and judge prompts will be
435provided in the supplementary material.

436 **6. Experimental setting/details.**

437 Answer: [Yes] Justification: Model identifiers, agent budgets, ablation configurations, and rubric
438 criteria are specified.

439 **7. Experiment statistical significance.**

440 Answer: [NA] Partial. Justification: Rubric judges run with three seeds; inter-rater reliability is
441 reported. The evaluation is a case study on representative bundles, not a statistical benchmark.

442 **8. Experiments compute resources.**

443 Answer: [Yes] Justification: API token constraints and execution bounds are documented.

444 **9. Research Ethics.**

445 Answer: [Yes] Justification: No human subjects or dual-use concerns apply.

446 **10. Broader impacts.**

447 Answer: [Yes] Justification: Highlights the positive impact of automating cumulative insight extrac-
448 tion while maintaining strict audit trails in regulated bioprocesses.

449 **A Architectural Audit: Supplementary Material**

450 **A.1 Auditor Configuration**

451 The Phase 1 architectural audit reported in Section 4.1 was performed using Gem-
452 ini 3.1 Pro accessed via opencode with file-system access to the FASSO repository at commit
453 1c28943513ba412af2bdf3335c419257c8697e63. Temperature was set to 0.1. The au-
454 dit was run in a fresh session with no shared context, no prior memory of the codebase, and no
455 communication with the implementation agent.

456 **A.2 Auditor System Prompt**

457 The complete system prompt provided to the auditor:

458 *"You are an independent reviewer for the FASSO project. The repository is on*
459 *disk. The paper draft is at paper/ read the .tex file there first to understand what*
460 *the authors claim. Then read the codebase to verify those claims against what is*
461 *actually built.*

462 *You are evaluating the work as a research and engineering artifact. You are not rat-*
463 *ing software craft (test coverage, code style, documentation quality, dependency*
464 *hygiene, CI configuration). You are rating the intellectual and engineering sub-*
465 *stance: how good is the idea, how faithfully does the implementation deliver on it,*
466 *and how well does the architecture fit its domain.*

467 *Rate the project on each of the following parameters, each out of 10. You de-*
468 *cide what each score means. Calibrate against the broader field of multi-agent*
469 *scientific reasoning systems, not against an idealized upper bound.*

470 **# PARAMETERS**

471 **1. Conceptual novelty.** *How original is the central idea relative to the existing*
472 *literature? Consider: - What does the paper claim is novel? - How does this*
473 *position against AI Co-Scientist, Robin, BioDiscoveryAgent, SciAgents, El Agente,*
474 *PiFlow, and the chem-eng MAS literature surveyed by Rupprecht et al. (2026)?*
475 *- Is the "world model-inspired" framing earning its weight or decorative? Are*
476 *the structural invariants a genuine reconceptualization or a relabeling of known*
477 *patterns (Pydantic validation, closed enums, pipelined artifacts)? - Distinguish*
478 *between novel synthesis (combining existing patterns in a non-obvious way) and*
479 *novel primitives (inventing new mechanisms).*

480 **2. Idea-to-execution fidelity.** *Do the three named structural invariants schema-*
481 *enforced citation, domain-scoped memory, observational-causal artifact separa-*
482 *tion actually exist in the code as described? Cite specific files and line ranges for*
483 *each invariant you verify, and explicitly note any claim type, code path, or layer*
484 *where the invariant is weaker than the paper suggests. Pay particular attention to:*

- Bypass paths: can a developer construct the protected object without triggering validation? - Coverage: is enforcement present for every claim type the paper names, or only some? - Layer consistency: the paper claims process_family is enforced "at three layers" verify this against the code. - Edge cases: "unknown" process families, fallback paths, concurrent paths.

3. Engineering depth. How well-engineered is the implementation relative to its ambition? Consider: - Does the multi-agent orchestration actually work as the paper describes, or does it work in a simplified or partial way? - Are the abstractions load-bearing (the system would break without them) or decorative (they could be removed without consequence)? - Is the architecture coherent across stages, or are there impedance mismatches between ingest, characterize, diagnose, and hypothesize? - Does the memory layer integrate cleanly, or does it sit awkwardly bolted on? - Are there parts of the system that exist only on paper or in stubs?

4. Domain effectiveness. How well does the system fit its stated problem (industrial fermentation hypothesis generation)? Consider: - Are the domain choices grounded in bioprocess analysis, or generic patterns dressed in domain vocabulary? - The kinetics / mass transfer / metabolic specialist decomposition is this how bioprocess engineers actually partition reasoning, or is it three roles chosen for architectural symmetry? - The closed vocabulary of process families is this enough to cover real industrial fermentation, or does it artificially constrain the problem? - Does the system handle the messy reality of fermentation data (Excel sheets, PDF reports, inconsistent units, scale changes across runs), or does it assume cleaner input than industry produces?

5. Paper-code alignment. Read the paper alongside the code. How well does the paper represent what the system actually does? - Are architectural diagrams faithful, or do they show idealized flows that don't quite match the runner? - Are the limitations the paper acknowledges the same limitations the code reveals, or has the paper missed some? - Are there things the code does that the paper doesn't mention? - Are there things the paper claims that the code does in a looser form? - Is the language calibrated (claims match evidence) or are there places where the framing is stronger than the implementation warrants?

OUTPUT FORMAT

Respond in exactly this structure. No prose outside it.

```
{ "conceptual_novelty": { "score": <1-10>, "evidence": "<two to three sentences with specific references to paper or comparable systems>", "key_strength": "<one sentence>", "key_weakness": "<one sentence must be a real weakness, not a token concession>" }, "idea_to_execution_fidelity": { "score": <1-10>, "evidence": "<two to three sentences with file:line citations for each invariant>", "bypass_paths_found": [ "<each bypass path you identified, with file:line; empty array only if you actively searched and found none>" ], "key_strength": "<one sentence>", "key_weakness": "<one sentence>" }, "engineering_depth": { "score": <1-10>, "evidence": "<two to three sentences citing specific design choices and load-bearing abstractions>", "decorative_or_stub_components": [ "<components that exist but are not load-bearing, with file paths; empty array only if you actively searched>" ], "key_strength": "<one sentence>", "key_weakness": "<one sentence>" }, "domain_effectiveness": { "score": <1-10>, "evidence": "<two to three sentences on how well domain-specific choices fit bioprocess analysis>", "key_strength": "<one sentence>", "key_weakness": "<one sentence>" }, "paper_code_alignment": { "score": <1-10>, "evidence": "<two to three sentences>", "places_paper_overclaims": [ "<each place where paper language is stronger than code warrants, with specific paper quote and code reference>" ], "places_paper_underclaims": [ "<each place where code does more than paper describes>" ], "key_strength": "<one sentence>", "key_weakness": "<one sentence>" }, "composite_score": <1-10, your overall rating>, "composite_rationale": "<two sentences explaining how you weighted the five parameters to arrive at the composite>", "headline_finding": "<one sentence summarizing the project's standing>", "would_recommend_for": "<which kind of venue this artifact fits>" }
```

The composite_score is your own judgment based on what matters most for the work's standing, not a mechanical mean of the five parameter scores.

545 The fields ‘bypass_paths_found’, ‘decorative_or_stub_components’,
 546 ‘places_paper_overclaims’, and ‘places_paper_underclaims’ are structural
 547 requirements. Empty arrays are only acceptable if you have actively searched
 548 for instances and genuinely found none. An empty array on a score below 9 is
 549 internally inconsistent at least one gap should be reported when the score is not
 550 maximal. State that you searched and found nothing, rather than leaving the field
 551 empty by default.

552 Cite specific file:line for code claims. Quote from the paper for paper claims.
 553 A reviewer reading your output should be able to verify every claim by clicking
 554 through to the cited location.

555 Be specific. Score honestly. Calibrate against the field, not against perfection.”

556 A.3 Audit Output

557 The auditor’s structured output is reproduced verbatim from FASSO_review.md:

558 **Headline Finding:** FASSO offers a highly novel structural approach to multi-agent
 559 scientific reasoning, enforcing invariants at the type level rather than via prompts.

560 **Overall Composite Score:** 9/10

561 **Composite Rationale:** FASSO successfully transposes world-model architecture
 562 into a strictly typed LLM pipeline, enforcing scientific grounding through schema
 563 validation and domain-scoped memory. It scores highly across the board due to the
 564 rare fidelity between its ambitious architectural claims and the rigorous Pydantic-
 565 enforced implementation.

566 **Recommendation:** A premier AI-for-Science conference (e.g., NeurIPS AI for Sci-
 567 ence workshop or ICLR) or a high-impact applied AI/cheminformatics journal.

568 1. Conceptual Novelty

569 Score: 8/10

- 570 • **Evidence:** The paper explicitly distances itself from textual plausibility systems
 571 like AI Co-Scientist and SciAgents, proposing a ‘world model-inspired’ frame-
 572 work based on perception, semantic compression, and causal hypotheses. The
 573 structural invariants, notably the closed-vocabulary `process_family` memory, move
 574 beyond typical multi-agent LLM systems into structured, domain-specific
 575 state transitions.
- 576 • **Key Strength:** Translating the world-model paradigm into strict semantic bound-
 577 aries and typed schema contracts, preventing typical LLM attention dilution.
- 578 • **Key Weakness:** Lacks actual learned forward dynamics, relying instead on se-
 579 mantic transition checks without physics-based numerical grounding.

580 2. Idea-to-Execution Fidelity

581 Score: 9/10

- 582 • **Evidence:** Invariant 1 is enforced via `@model_validator`
 583 in `src/fermdocs_diagnose/schema.py:120` and
 584 `src/fermdocs_hypothesis/schema.py:271`. Invariant 2 is strictly
 585 checked in `src/fermdocs_memory/base.py:101`, throwing a `ValueError`
 586 if no process family is provided. Invariant 3 is maintained by typed schemas
 587 that cleanly separate observational limits (`AnalysisClaim.kind`) from causal
 588 predictions (`HypothesisFull`).
- 589 • **Bypass Paths Found:**
 590 – `src/fermdocs_hypothesis/agents/synthesizer.py:510` –
 591 The Synthesizer agent automatically backfills citations from the
 592 `citation_universe` if the LLM drops them, satisfying the schema
 593 but circumventing strict LLM citation discipline.
- 594 • **Key Strength:** The three named invariants are directly built into load-bearing,
 595 type-checked Pydantic contracts rather than relying on prompt engineering.
- 596 • **Key Weakness:** The safety-net backfill in the synthesizer weakens the cognitive
 597 enforcement of Invariant 1 on the LLM itself.

598 3. Engineering Depth

599 Score: 9/10

- 600 • **Evidence:** The architecture is extremely robust, utilizing highly struc-
 601 tured state machines and typed schemas. The `JudgeView` in

602 `src/fermdocs_hypothesis/schema.py:330` brilliantly enforces asymmetric information by hiding previous debate histories to prevent collusion and
603 sycophancy.
604
605 • **Decorative or Stub Components:**
606 – `src/fermdocs_hypothesis/schema.py:356` - `HumanInputRecord` is
607 a stub for future HITL ('Empty in v0').
608 • **Key Strength:** The asymmetric information design and rigid JSON schemas
609 prevent the system from drifting into the typical sycophantic failure modes of
610 multi-agent debate.
611 • **Key Weakness:** The pipeline's intense rigidity around schema structures may
612 make it brittle to novel or malformed fermentation run data.

613 **4. Domain Effectiveness**
614 **Score: 8/10**

615 • **Evidence:** The system decomposes tasks into Kinetics, Mass Transfer, and
616 Metabolic specialists, aligning deeply with industrial bioprocess paradigms. The
617 closed-vocabulary `process_family` restricts cross-strain hallucinations, which
618 is a major unmet need highlighted by the Rupprecht review.
619 • **Key Strength:** The `process_family` scoping structurally prevents cross-
620 domain contamination, grounding the memory securely in domain reality.
621 • **Key Weakness:** The reliance on predefined anomaly and finding vocabularies
622 limits its ability to articulate biological mechanisms completely outside the estab-
623 lished ontology.

624 **5. Paper-Code Alignment**
625 **Score: 9/10**

626 • **Evidence:** The paper is exceptionally transparent, explicitly acknowledging its
627 limitation in lacking numerical forward simulations (the 'JEPA gap'). The architec-
628 tural claims align accurately with the codebase's Pydantic models and memory
629 protocols.
630 • **Places Paper Overclaims:**
631 – The paper claims making unlinked claims 'impossible to construct' (Section 1),
632 but `src/fermdocs_hypothesis/agents/synthesizer.py:510` auto-
633 backfills citations, meaning the agent can technically emit unlinked claims that
634 are quietly fixed.
635 • **Places Paper Underclaims:**
636 – The codebase implements sophisticated confidence downgrading (`provenance_downgraded` in
637 `src/fermdocs_diagnose/schema.py:118`) and complex seed topic
638 ranking (`src/fermdocs_hypothesis/seed_topic_extractor.py`) not
639 detailed in the paper.
640 • **Key Strength:** Honest assessment of system limitations, particularly acknowl-
641 edging its step toward world models without claiming full learned dynamics.
642 • **Key Weakness:** The rhetoric around 'impossible to construct' is slightly over-
643 stated given the automated backfill safety-nets.
644